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Beliefs and practices of Anganwadi and private preschool teachers towards emergent literacy

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ABSTRACT
This cross-sectional observational study explored the nature of beliefs and practices regarding emergent literacy among 123 preschool teachers across two school types (private school and Anganwadi) in the Indian context. The findings revealed that teachers in both school types expressed strong beliefs in the role of emergent literacy in preschool and associated practices. Beliefs varied based on school type and teachers' educational qualifications. The self-reported practices and quantity of time taken for emergent literacy activities were not influenced by the difference in school types, teachers' experience, educational qualification or teacher age. The study details the implications for teacher training.

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Q1 Introduction

The National Educational Policy (NEP 2020) in India recommends early childhood education as a foundational step in school education, recognising that early experiences shape children's academic, language, cognitive and socio-emotional development. Yet, national assessments reveal persistent gaps in foundational learning. The Annual Status of Education Report (ASER 2022) documented a decline in reading and arithmetic skills among children in Classes 3 and 5, with performance dropping to pre-2012 levels across both government and private schools. This pattern reflects a persistent challenge Q2 noted in previous decades (Chatterjee and Robitaille 2018). Encouragingly, the most recent ASER (2024) reported improvement in Class 3 outcomes, with particularly notable gains in government schools. These national trends suggest that strengthening foundational skills in the preschool years may be key to sustaining improvements in later grades.

High-quality preschool practices can mitigate later learning gaps. In particular, preschool experiences with emergent literacy practices (i.e. foundational skills, knowledge,

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and attitudes for conventional forms of reading and writing) increase the probability of a child's long-term academic success (Chopra and Khanna 2019). Recommendations highlighted in the NEP (2020) underscore an urgency for universal access to high-quality pre-primary school programmes that teach foundational literacy skills. The NEP (2020) emphasises teacher training as a key policy component. However, as noted by Gupta (2022), policies should be aligned with the existing cultural hybridity of Indian beliefs for preschool education. This study examines preschool teachers' self-reported beliefs and practices regarding emergent literacy in the Udupi district of Karnataka, South India, across two settings: Anganwadi and private schools. The findings aim to inform curriculum development and teacher training initiatives in alignment with national policy goals.

Early education and emergent literacy: the Indian context

Emergent literacy skills that support the acquisition of reading and writing can be broadly classified as code-related (i.e. phonological awareness, print knowledge) and meaning-related (i.e. oral language and comprehension) skills (Justice, Sofka, and McGinty 2007; Lonigan et al. 2013; Rohde 2015; Sulzby and Teale 1991; Whitehurst and Lonigan 1998). The emergent literacy perspective considers literacy acquisition as a developmental continuum supported by experiences within rich emergent literacy environments at home and school (Mol and Bus 2011). Robust literature points towards best practices in preschool classrooms to promote emergent literacy. Most research related to best practices is predominantly from a Western context and recommends both code- and meaning-related skills concurrent with explicit instruction and scaffolding (Goldfeld et al. 2022; Joo et al. 2020). Other characteristics associated with best-practice instruction include a small group of three to five students and high-intensity instruction (e.g. four to five sessions per week, 30 min per session, 3–6 months duration).

Although limited, research conducted in India examining emergent literacy programmes leans towards best practices adapted from Western contexts. This trend is unsurprising, given that most investigations are situated in English medium schools, where the language of instruction is English. To fully situate this research, it is important to consider the main types of preschool provision in India, where pre-primary education is offered through Anganwadi and private school types, providing instruction in English or the state language.

School types: Anganwadi and private

Anganwadi is a government-provided child-care and mother-care development national programme under the Integrated Child Development Scheme (ICDS) in India. As part of the public health care system, it is involved in pre-primary education, health and nutrition education, supplementary nutrition, immunisation, health check-up and referrals. It is reported that approximately 1,363,000 Anganwadi centres covering 88 million children aged below 6 years of age are available in India (Ministry of Women & Child Development, Government of India 2018). Usually, the Anganwadi school types are rented places or residences in the neighbourhood and have high pupil–teacher ratios. They are based in

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the community where every Anganwadi worker reaches around 250 households (Subramanian 2019).

Private schools are typically managed by private individuals, trusts or organisations. Unlike Anganwadi centres, private schools are fee-based and are more likely to offer instruction in English, particularly in urban areas. Pre-primary classrooms in private schools tend to emphasise early academic skills, and pedagogy is often rigid and oriented towards rote memorisation, leaving less room for developmentally appropriate, play-based and activity-based learning (Kaul et al. 2017). With this structural context in mind, the following section describes research examining emergent literacy practices in Indian preschools.

Evidence from Indian preschools

In a study among 28 preschool teachers from 10 English medium pre-primary schools in Mysore city, Khurana and Rao (2011) reported on the emergent literacy experiences of children. On a 'questionnaire for teachers' and 'questionnaire on books', it was documented that the preschoolers were exposed to various emergent literacy activities, including storybook reading, letter knowledge, phonological awareness, concepts of print and oral language. Most preschools in the sample delivered high-quality, child-friendly books with age-appropriate language and illustrations, resulting in rich literacy.

Building on this line of inquiry, more recent studies have examined the application of structured literacy curricula and teacher training in English medium schools. Zahedi et al. (2022) reported on the results of a strategic plan to improve foundational reading skills of kindergarten children in a PreK-10 private English medium school regulated under one of the national educational boards (Indian Certificate of Secondary Education Examination – ICSE) located in Mumbai, India. The school implemented a curriculum that included the use of child-centred activities, direct instruction on code-related aspects (e.g. Jolly Phonics) and meaning-related aspects of emergent literacy (e.g. HighScope). They also put into place a comprehensive approach for teacher training and the involvement of parents. Literacy class occurred five times a week for 30 min per class, and the plan was to increase the amount of time per class. The authors attribute the success of the strategic plan implementation to the integrated approach undertaken.

While Zahedi et al. (2022) highlighted the benefits of an integrated, systemic approach, Shenoy, Iyer, and Zahedi (2024) focused more specifically on the role of phonological awareness in supporting early literacy development. They examined literacy outcomes following implementation of a preschool curriculum with a focus on code-related skills (e.g. a multi-sensory approach to sound-symbol correspondence; sound segmentation and blending). Instruction was small group, high intensity (30 min per day; 5 days per week). Professional development sessions for teachers were also conducted. Outcomes ($n = 627$ students) indicated that at-risk students who received this instruction performed better on literacy skills in kindergarten. The authors recommended the inclusion of systematic teaching of phonological awareness skills in Indian preschools to improve early literacy skills and mitigate against later reading challenges.

While these findings show the potential of structured curricula and teacher training, generalisation to other school contexts in India is limited due to the variance in Anganwadi and private schools described above. Furthermore, there are no precise programme

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standards in place for preschool or kindergarten classrooms in India (Hegde and Cassidy 2009), and teachers in both school types often have limited or inconsistent training in early childhood education. In such contexts, teachers' beliefs and attitudes towards emergent literacy become especially important, as these shape how instructional practices are enacted in the classroom.

Teachers' beliefs and attitudes: connection to practice

Teachers' knowledge and beliefs influence their classroom behaviour and practice (Lopes and Fernandes 2009). Cadime et al. (2025) explored the relationship between preschool teachers' beliefs and practices regarding emergent literacy development in Portugal. They found that beliefs that emergent literacy is a social, community-based and culturally defined process were associated with a greater frequency of practices that promote teaching both code-related and meaning-related aspects of emergent literacy, as well as engaging in meaningful, integrated and child-centred approaches to literacy development. The study also emphasised the importance of continuous training in fostering holistic beliefs and evidence-based practices among preschool teachers. Lynch and Owston (2015) investigated Canadian preschool teachers' beliefs about language and literacy teaching and their alignment with research-based best practices. Results indicated that the teachers were uncertain in their beliefs about language and literacy practices, in particular code-related knowledge. The authors recommend the need for supporting preschool teachers using research-informed professional development training programmes, focussing on both code-related and meaning-related knowledge, for language and literacy development. On the other hand, Cash et al. (2015) did not find the beliefs of teachers in the US to be predicative of emergent literacy outcomes, but their knowledge of language and literacy positively predicted the children's gains in expressive vocabulary and print knowledge. The study emphasises the need for teacher training programmes focussing on teachers' knowledge about language and literacy development in the early childhood period.

While most guidelines related to early childhood practices are adopted from Western contexts, preschool teachers' beliefs and practices with regard to emergent literacy may vary depending on cultural influences and structural constraints. McMullen et al. (2005) examined pre-primary professionals in five countries (US, China, Taiwan, Korea and Turkey) to explore how their philosophies aligned with or differed from the US defined appropriate practices. They found that while Western ideas are spread globally, their application is shaped by cultural norms (e.g. individualism vs. collectivism), structural constraints (e.g. adult–child ratios) and local professional values (e.g. tensions between professional values of child-centred practices and parent expectations). In the first validated study in Uzbekistan, Muminova (2020) examined preschool teachers' beliefs and practices related to emergent literacy. Findings showed a significant gap between research-based concepts and current teacher beliefs/practices. Increasing knowledge and beliefs through teacher training along with structural, systemic reforms were identified to promote effective change.

In the Indian context, Anjana and Rao (2019) surveyed 48 preschool teachers from 10 central board private schools (a national-level education system) in Kerala, a southern state of India, about their beliefs regarding emergent literacy. Results indicated that

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teachers had positive beliefs about their role in supporting emergent literacy irrespective of their years of experience and education. Specifically, teachers had a strong agreement on the questions about belief in code-related skills (phonics, rhyming, print knowledge and alphabetic principle), the role of school and the role of parents, suggesting stronger beliefs and positive knowledge in these domains. However, their belief regarding the influence of oral language abilities on literacy was low. The teachers also showed limited knowledge about early identification of problems and teaching strategies, suggesting the need to focus on these aspects during in-service training programmes. The study did not report on teacher practices in the classroom.

Hegde and Cassidy (2009) examined the beliefs, stated practices and actual practices regarding developmentally appropriate practices among 40 kindergarten teachers from the middle and upper socio-economic groups in an urban city in Mumbai, India, within private school types. The Teacher Belief Scale, Instructional Activities Scale and Classroom Practice Inventory reported no difference across the two grade levels (upper and lower kindergarten). The actual practices were found to be more congruent with stated practices than with the stated beliefs. The stated beliefs were more developmentally appropriate than the stated and actual practices. Teachers with developmentally appropriate beliefs and stated practices also exhibited developmentally appropriate actual practices (e.g. use of building blocks instead of rote learning). The classroom group size was found to predict the teacher's actual practice compared with other variables such as the teacher's education, experience, beliefs or classroom composition.

These findings suggest that while Indian teachers may hold generally positive beliefs about emergent literacy, gaps persist. Understanding teachers' beliefs and practices across different school types is a critical step towards designing contextually responsive training initiatives. This study addresses that gap by examining teachers' beliefs and practices in Anganwadi and private preschool settings in Udupi district, Karnataka.

Summary and study aims

In sum, there is no single solution for addressing the diversity in India's pre-primary schools. More knowledge of teachers' stated beliefs and classroom practices across school types and regions is needed. Existing evidence from English medium schools suggests that code-related and meaning-related instruction can support early literacy development. However, the limited studies conducted in India are often restricted to a single school type or small sample size. Given the diversity of preschool provision, particularly between Anganwadi and private schools, findings from one setting cannot be readily generalised to others.

The present study addresses this gap by investigating preschool teachers' self-reported beliefs and practices regarding emergent literacy in the Udupi district, Karnataka, South India. Udupi is linguistically and socially diverse, with commonly spoken languages including Tulu, Konkani, Kannada and Beary, and residents from varied socio-economic and occupational backgrounds. Specifically, this study examines whether teachers' emergent literacy beliefs and practices differ by school type (Anganwadi vs. private), as well as by teacher characteristics such as educational qualifications, years of experience and age.

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Materials and methods

A cross-sectional observational study design was employed for the present study. The researchers initially obtained prior approval from the Institutional Ethics Committee (IEC-378/2021) and registration in the Clinical Trial Registry-India (CTRI/2021/08/035639). After obtaining permission from the district authorities, the schools in the district were initially identified based on the list of private schools and Anganwadi institutions obtained from the district education officer. Permission to conduct the studies in respective school types was obtained by formal communication over phone or e-mail, followed by a meeting with the school authorities to inform them about the study details.

Participants

The preschool teachers from the two school types (Anganwadi and Private school types) were the study participants. Through convenience sampling, 123 study participants (63 Anganwadi preschool teachers and 60 private preschool teachers) were recruited from 63 Anganwadi and 12 private schools within the Udupi district, Karnataka state, in South India. Preschool teachers with a minimum of 2 years of teaching experience and teaching young children aged 3–6 were included in the study. Untrained teaching assistants or students undergoing teacher training, or teachers involved with training children with special needs, were excluded.

The study participants fulfilling the selection criteria completed a questionnaire on emergent literacy (Appendix A). The demographic details, such as age, gender, education, designation in school, years of experience, school type, number of students enrolled in the class, age group of children taught, mode of teaching and medium of instruction, were obtained using the questionnaire. The next section provides details of the questionnaire preparation, data collection and data analysis.

Questionnaire preparation and validation

Initially, the items for the questionnaire were prepared to comprise both closed-ended and open-ended questions from the existing research (Burts et al. 2000; Sandvik, Van Daal, and Adèr 2014; Seefeldt 2004). After the first author identified the items within the domains (personal information, classroom details, stated beliefs and practices regarding emergent literacy), the two experienced speech-language pathologists (SLPs) (>15 years of experience) individually reviewed each item for cultural appropriateness, clarity, representativeness, word choice and meaning. Only items confirmed appropriate by all three researchers were included in the study questionnaire.

The researchers consulted five preschool teachers with at least 5 years of experience working with preschool children and five SLPs with a minimum of 3 years of experience in dealing with childhood language conditions. The raters from an SLP background were considered as they were familiar and knowledgeable about the study context and the emergent literacy practices followed in the preschool context. The SLPs play a major role in identifying children at risk of developing literacy-related learning difficulties and assisting teachers in providing support for children with language-literacy concerns.

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They help in creating research-to-practice relationships that assist teachers in effectively using print referencing in their classrooms (Justice et al. 2009). They are often one of the health professionals who are the initial point of referral. SLPs also help preschool teachers become skilled facilitators by helping them gain knowledge of how young children acquire print knowledge, language development, the phonological system, vocabulary, sentence structure and comprehension as integral members of the school team (Powell 2018). The experts were asked to validate each questionnaire item as either relevant, irrelevant or needing revision specific to the Indian context. The items confirmed as valid by all of the investigators were included in the study questionnaire. Some notable changes include the following: (1) the terminology 'letter knowledge' was rephrased as '*Akshara*/letter knowledge'; (2) literacy-related activities according to the Indian context were included (e.g. *Akshara* grids, dice, word cards, *Ganitmala*, blocks, beads, puzzles and board games or having imaginary play with dolls, masks, puppets) and (3) 'writing down of stories told by children' was rephrased to 'story mapping by drawing the characters of the story that interest the children'.

After the validation process, the questionnaire was translated and made available in standard Kannada (the official and most spoken Kannada dialect) using a forward–backward translation procedure. Initially, in the forward translation, the questionnaire was translated from English to Kannada by a bilingual translator, proficient in both English and Kannada. Two persons read the translated questionnaire for word choice, sentence structure, clarity and meaning. This translated questionnaire was given to another bilingual translator to carry out the backward translation, Kannada to English. The final questionnaire (Appendix A), after validation and translation, included the following two sections.

Section A – Demographic details of study participants

This section includes details related to the personal information of teachers and classroom setup.

Section B – Beliefs and practices regarding emergent literacy section

This section comprised three subsections (belief, practice and quantity of time) with the survey questionnaire items adapted from the Preschool Literacy Survey (Sandvik, Van Daal, and Adèr 2014) with the author's permission. Additional questions were adapted from other studies (Burts et al. 2000; Seefeldt 2004).

The Belief section of the survey included items within the following categories: Role of the teacher (17 statements); Role of the pre-primary school (11 statements) and Consistency with current research (8 statements). The Practices section of the survey included items within the following categories: Quality of shared book reading (14 statements); Concepts of book and print (7 statements); Literacy in play (5 statements); Phonological awareness (6 statements); Letter knowledge (5 statements) and Emerging reading and writing (9 statements). The Quantity of Time section consisted of eight statements related to time spent on literacy-related activities. These questions are annotated in Appendix A.

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Data collection

The first author recruited the preschool teachers fulfilling the selection criteria after obtaining written informed consent. A face-to-face interview was carried out to obtain information regarding the demographic details (Section A of the questionnaire). Teachers were also asked to report practices and strategies that they used to promote literacy so that their responses to the practice section of the questionnaire could be contextualised.

Next, the teachers completed the self-rated questionnaire survey (Section B of the questionnaire), which took approximately 20–30 min per participant. The questionnaire-based section consisted of 90 items, where the participants were asked to rate each item on a six-point Likert-like scale.

In the Belief section (36 statements), the participants were instructed to score the items on a six-point Likert scale, with one indicating 'strongly disagree' and six indicating 'strongly agree'. In the Practice section (46 statements), the participants rated themselves from 1 to 6, with one indicating 'never' and six indicating 'always'. In the last section on the quantity of time they spend each day with children towards early learning (8 statements), the participants rated themselves over a six-point Likert scale, with one indicating 'zero' and six indicating '20 or more minutes'.

Data analysis

For the survey questionnaire responses, each participant's response was computed for each subsection, and then, the total score for each subsection was obtained. Descriptive statistics were used to compute and analyse all the variables using Jamovi 2.0 (The Jamovi Project 2021). A two-sample independent *t*-test was used to check the statistical significance of the average scores obtained in beliefs, practices and quantity of time across the dichotomous categories associated with the demographic variables. The level of significance was set at 5% throughout the study. Responses to the informal question about practices (i.e. activities and strategies used to promote literacy in the classroom) were classified according to the practice categories listed in the survey questionnaire.

Results

Section A: Demographic characteristics of study participants

A total of 123 teachers aged between 23 and 59 years (M = 42.9; SD = 8.13) were surveyed (see Table 1). Approximately one-half of the teachers worked in either Anganwadi (n = 63) or private schools (n = 60). Their work experience ranged from 3 to 35 years, with 38 having more than 20 years of working experience. The majority (n = 75) reported basic education qualifications (i.e. 10 or 12 years of education with Anganwadi training/Diploma in Early Education/Teacher training in Early Education/Nursery Teacher Training/ECCE/Montessori training/Early Childhood Care and Education). Forty-eight teachers had higher qualifications (i.e. graduation/bachelor's or post-graduation/master's degree). The teachers dealt with children between 3 and 6 years of age, with the number of children taught by each teacher varying between 5 and 94 (total

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Table 1. Demographic details of the study participants.

Variables	Number of participants (*n*)	Percentage
Gender		
Female	123	100
Male	0	–
Age		
23–41 years	46	37.39
42–59 years	77	62.60
	Educational qualification	
Basic	***qualification***	
75	60.97	
	Anganwadi training	
SSLC and		
17	13.82	
PUC and	Anganwadi training	

(*Continued*)

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number of students as reported). The common languages spoken by teachers during the class sessions were Kannada, English and Tulu.

Section B: Teachers' beliefs, practices and quantity of time

A. Overall beliefs, practices and quantity of time

As indicated in Table 2, mean responses for beliefs were strong for both Anganwadi and private school teachers; in that of the maximum score of 216, the Anganwadi and private school teachers scored 201.59 and 195.58, respectively. The results indicated a statistically significant difference in mean belief scores based on school types, with Anganwadi teachers having higher mean belief scores (M = 201.59 (95% CI: [197.9, 205.3]); SD = 14.66) as compared with private school teachers (M = 195.58 (95% CI: [191, 200.2]); SD = 17.90), (t (121) = 2.03, p-value = 0.04).

Mean responses for practices were strong for both Anganwadi and private school teachers, with a maximum score of 276; teachers scored 237 and 245, respectively. There were no statistically significant differences in the mean scores for practices between Anganwadi (M = 237 (95% CI: [229.2, 243.9]); SD = 29.3) and private school (M = 245 (95% CI: [239.6, 250.4]); SD = 20.8) teachers.

Mean responses for the quantity of time were relatively high for both Anganwadi and private school teachers, with a maximum score of 48; teachers scored 40.57 and 39.25, respectively. There was no statistically significant difference in the mean scores for the

Table 2. Beliefs, practices, and quantity of time, based on school setup, experience, educational qualification and age.

Characteristic	Beliefs (Max score: 216) Mean (SD)	Beliefs Test statistic (*p*-value)	Practices (Max score: 276) Mean (SD)	Practices Test statistic (*p*-value)	Quantity of time (Max score: 48) Mean (SD)	Quantity of time Test statistic (*p*-value)
School setup						
Anganwadi	201.59 (14.66)	2.03 (0.04) *	237 (29.3)	1.8 (0.07) NS	40.57 (6.67)	1.1 (0.28) NS
Private	195.58 (17.90)		245 (20.8)		39.25 (6.89)	
Years of experience						
Less than 19 years	198.39 (17.13)	0.22 (0.82) NS	241.85 (25.83)	0.65 (0.52) NS	40.33 (6.59)	0.83 (0.40) NS
More than 20 years	199.08 (15.70)		238.74 (25.83)		39.27 (7.09)	
Educational qualification						
Basic qualification	196.37 (18.76)	1.64 (0.10) NS	234.68 (29.17)	2.8 (0.06) NS	39.71 (7.04)	0.37 (0.71) NS
Higher qualification	201.22 (13.31)		247.38 (19.47)		40.17 (6.54)	
Age						
Less than 41 years	199 (15.8)	0.38 (0.71) NS	243 (26)	0.7 (0.49) NS	40.1 (6.99)	0.20 (0.84) NS
More than 42 years	198 (17.1)		239 (25.7)		39.8 (6.70)	

NS, non-significant; *$p < 0.05$.

quantity of time between Anganwadi (M = 40.57 (95% CI: [38.9, 42.3]); SD = 6.67) and private school (M = 39.25 (95% CI: [37.5, 41.1]); SD = 6.89) teachers.

In terms of years of experience, no statistically significant group differences between lower and higher years of experience were noted for the belief and quantity of time. In the context of practice aspects, there were no statistically significant group differences across the two levels of experience, (t (121) = 0.65, p-value = 0.52), despite study participants with lesser years of experience having a slightly higher mean score (M = 241.85 (95% CI: [236, 247.8]); SD = 25.83) than the higher experience group (M = 238.74 (95% CI: [231.2, 246.3]); SD = 25.83) along with highly overlapping confidence intervals.

On the comparison of the educational qualifications of the teachers, there was no statistically significant group difference (t (121) = 1.64, p-value = 0.10), even though a higher mean score in the beliefs section was noted for teachers with higher educational qualifications (M = 201.22; (95% CI: [197.7, 204.7]); SD = 13.31) than for the teachers with basic qualifications (M = 196.37 (95% CI: [191.7, 201]); SD = 18.76). No significant group differences based on educational qualification were noted for the total mean scores for practices and quantity of time.

In terms of the age of the participants, no statistically significant differences were noted in the mean scores of beliefs and quantity of time. For practices, there was no statistically significant group difference based on age (t (121) = 0.7, p-value = 0.49), despite the younger participants having higher mean scores (M = 243 (95% CI: [235, 250]); SD = 26) than the older participants (M = 239 (95% CI: [233.6, 245.3]); SD = 25.7).

B. Beliefs of teachers toward emergent literacy

This section indicates the teachers' perspectives on three subdomains of beliefs. As indicated in Table 3, no statistically significant group difference was noted for the 'role of

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Table 3. Preschool teachers' beliefs on emergent literacy across school setup, experience, educational qualification and age.

Beliefs	Role of preschool teacher (Max score: 102)		Role of preschool (Max score: 66)		Consistency with current research (Max score: 48)	
	Mean (SD)	Test statistic (*p*-value)	Mean (SD)	Test statistic (*p*-value)	Mean (SD)	Test statistic (*p*-value)
School setup						
Anganwadi	97 (6.25)	2.6 (0.01) *	60.13 (6.45)	0.19 (0.85) NS	44.46 (3.61)	2.81 (0.06) NS
Private	93.30 (9.25)		60.33 (5.76)		41.95 (6.05)	
Years of experience						
Less than 19 years	95.13 (8.65)	0.11 (0.91) NS	60.19 (6.08)	0.07 (0.94) NS	43.06 (5.43)	0.47 (0.64) NS
More than 20 years	95.29 (7.04)		60.28 (6.19)		43.51 (4.52)	
Educational qualification						
Basic qualification	94.58 (8.69)	0.89 (0.37) NS	58.71 (6.94)	3.02 (0.03) *	43.08 (5.20)	0.36 (0.72) NS
Higher qualification	95.88 (7.26)		61.93 (4.45)		43.41 (5.001)	
Age						
Less than 41 years	95.6 (8.15)	0.43 (0.66) NS	60.7 (5.56)	0.59 (0.55) NS	43.1 (5.57)	0.18 (0.86) NS
More than 42 years	94.9 (8.03)		60 (6.42)		43.3 (4.82)	

NS, non-significant; *$p < 0.05$.

preschool teacher' section in terms of years of teaching experience, educational qualification and participants' age. Across the school setup, a statistically significant total mean score for belief was noted for Anganwadi teachers (M = 97 (95% CI: [95.4, 98.6]); SD = 6.25) compared with the private school teachers (M = 93.30 (95% CI: [90.9, 95.7]); SD = 9.25), (t (121) = 2.6, p-value = 0.01).

For the section on 'role of preschool', no statistically significant difference in the mean scores was noted in terms of schools set up, years of teaching experience and participants' age. Based on educational qualification, the results indicated a statistically significant higher total mean score for participants with higher educational qualifications (M = 61.93 (95% CI: [60.8, 63.1]); SD = 4.45), as compared to those with lower educational qualifications (M = 58.71(95% CI: [57, 60.4]); SD = 6.94), (t (121) = 3.02, p-value = 0.03).

No group differences between Anganwadi and private school teachers on the consistency with current research score were noted. No group differences between years of teaching experience, educational qualifications and participants' age were observed in the present study.

C. Practices of teachers towards emergent literacy

This section indicates the teachers' perspectives on six subdomains of practices (i.e. quality of shared book reading, concepts of books and print, literacy in play, phonological awareness, letter/*Akshara* knowledge, emerging reading and writing) across school setup, experience, educational qualification and age.

As indicated in Table 4, teachers with higher educational qualifications reported statistically significantly greater use of shared book reading (M = 79.15 (95% CI: [77.8, 80.5]); SD = 5.16, t (121) = 3.59, p-value < 0.001), concepts of books and prints (M = 36.89 (95% CI: [35.7, 38.1]); SD = 4.61, t (121) = 2.05, p-value = 0.04), literacy in play (M = 25.27

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Table 4. Preschool teachers' practices related to emergent literacy across school setup, experience, educational qualification and age.

Practices	Quality of shared book reading (Max score: 84)		Concepts of books and print (Max score: 42)		Literacy in play (Max score: 30)		Phonological awareness (Max score: 36)		Letter/*Akshara* knowledge (Max score: 30)		Emerging reading and writing (Max score: 54)	
	Mean (SD)	Test statistic (*p*-value)	Mean (SD)	Test statistic (*p*-value)	Mean (SD)	Test statistic (*p*-value)	Mean (SD)	Test statistic (*p*-value)	Mean (SD)	Test statistic (*p*-value)	Mean (SD)	Test statistic (*p*-value)
School type												
Anganwadi	75.54 (8.06)	1.68 (0.096) NS	34.54 (6.85)	2.56 (0.012)*	24.28 (4.97)	0.27 (0.788) NS	30.11 (6.42)	2.12 (0.036) *	27.60 (3.39)	0.69 (0.49)NS	44.48 (7.46)	0.78 (0.436) NS
Private	77.83 (7.04)		37.12 (3.79)		24.52 (4.52)		32.12 (3.60)		27.97 (2.36)		45.43 (5.99)	
Years of experience												
Less than 19 years	77.22 (7.78)	1.43 (0.29) NS	36.17 (5.85)	0.93 (0.36) NS	24.49 (4.75)	0.26 (0.79) NS	31.04 (5.91)	0.13 (0.89) NS	28 (2.84)	1.06 (0.29) NS	44.93 (7.03)	0.018 (0.98) NS
More than 20 years	75.74 (7.40)		35.19 (5.44)		24.25 (4.77)		31.17 (4.24)		27.42 (3.05)		44.96 (6.41)	
Educational qualification												
Basic qualification	74.43 (8.77)	3.59 (<0.001) *	34.81 (6.39)	2.05 (0.043) *	23.61 (5.37)	1.99 (0.05) *	29.93 (6.27)	2.60 (0.01) *	27.58 (3.311)	0.78 (0.43) *	44.29 (7.58)	1.13 (0.26)
Higher qualification	79.15 (5.16)		36.89 (4.61)		25.27 (3.77)		32.38 (3.61)		28 (2.43)		45.67 (5.72)	
Age												
Less than 41 years	77.5 (7.05)	0.9438 (0.347) NS	36.3 (5.15)	0.76 (0.447) NS	24.5 (4.76)	0.26 (0.794) NS	31.1 (6.22)	0.031 (0.975) NS	28.2 (2.32)	1.22 (0.225) NS	45.1 (7.64)	0.15 (0.88)
More than 42 years	76.2 (7.97)		35.5 (6.01)		24.3 (4.76)		31.1 (4.74)		27.5 (3.23)		44.9 (6.26)	

NS, non-significant; *$p < 0.05$.

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(95% CI: [24.3, 26.3]); SD = 3.77, t (121) = 1.99, p-value = 0.05) and phonological awareness activities (M = 32.38 (95% CI: [31.4, 33.3]); SD = 3.61, t (121) = 2.6, p-value = 0.01), than teachers with lower educational qualifications.

Private school teachers reported the use of concepts of book and print activities (M = 37.12 (95% CI: [36.1, 38.1]); SD = 3.79) and phonological awareness activities (M = 32.12 (95% CI: [31.2, 33]); SD = 3.60) with greater frequency than Anganwadi teachers, i.e. concepts of book and print (M = 34.54 (95% CI: [32.8, 36.3]); SD = 6.85) and phonological awareness (M = 30.11 (95% CI: [28.5, 31.7]); SD = 6.42). These differences were statistically significant for both concepts of book and print (t (121) = 2.56, p-value = 0.01) and phonological awareness (t (121) = 2.12, p-value = 0.04).

No significant group differences regarding school setup, years of experience, educational qualification and participants' age were noted for practices related to *Akshara* knowledge and emergent reading and writing.

As indicated in Table 5, teachers responded with a variety of activities they considered to promote literacy in the classroom. Emergent literacy activities reported by the majority included engaging children in literacy through play and the use of shared book reading. Fewer teachers reported activities and strategies for explicit teaching of print and phonological awareness (e.g. rhyming, book and print concepts, letter/*Akshara* knowledge).

D. Quantity of time spent by teachers on emergent literacy activities

This section gives an approximate time spent by preschool teachers on literacy-related activities each day (Table 6). It was seen that more than 20 min was taken daily for doing various emergent literacy activities including helping children write (63.41%), doing alphabet activities with children (59.35%), talking about the relationship between letters and sounds (47.15%), helping children learn letters (46.34%), reading and writing through play (41.46%), story mapping (40.65%), reading aloud (37.4%) and listening to children while reading or pretending to read (32.52%). As indicated earlier in Table 3, there was no statistically significant group difference between school types, years of experience, educational qualification or the age of the teachers for the total mean scores on the quantity of time taken.

Table 5. Categories and subcategories derived from teachers' responses to activities used to promote literacy.

Topic	Category	Subcategory	Activities
Activities that enhance the participation of children in the learning process	Emergent literacy teaching activities	Quality of shared book reading	Using pictures, actions and charts concurrent with books
		Concepts of books and printing	Drawing and colouring with literacy materials
		Literacy in play	Using art, dance and toys related to literacy materials
		Phonological awareness	Rhyming activities
		Letter/*Akshara* knowledge	Writing on lines, tracing letters in sand and water
		Emerging reading and writing	Counting, oral language drill activities, use of *Shalapoorva Shiksha*, use of *Nali-Kali* programme, reading
		Others	Showing videos on the phone, engaging in classroom competitions, storytelling

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Table 6. Quantity of time daily spent by preschool teachers on emergent literacy activities.

Activities	Zero *n* (%)	0–5 min *n* (%)	5–10 min *n* (%)	10–15 min *n* (%)	15–20 min *n* (%)	20 or more minutes *n* (%)
I read aloud to children in preschool						
Anganwadi	1(1.59)	0	5(7.93)	12(19.04)	21(33.33)	24(38.09)
Private	0	2(3.33)	14(23.33)	14(23.33)	9(15)	22(36.67)
Total	**1 (0.81)**	**2 (1.63)**	**19 (15.44)**	**26(21.14)**	**30 (24.39)**	**46 (37.4)**
I listen to children read or pretend to read aloud						
Anganwadi	4(6.35)	2(3.17)	10(15.87)	13(20.63)	15(23.81)	19(30.16)
Private	1(1.67)	5(8.33)	15(25)	7(11.67)	11(18.33)	21(35)
Total	**5 (4.06)**	**7 (5.7)**	**25 (20.32)**	**20 (16.26)**	**26 (21.14)**	**40 (32.52)**
I map the story by drawing the characters of the story that are of interest to the children						
Anganwadi	0	3(4.76)	4(6.35)	6(9.52)	20(31.74)	30(47.61)
Private	0	2(3.33)	10(16.67)	12(20)	16(26.67)	20(33.33)
Total	**0**	**5 (4.1)**	**14 (11.38)**	**18 (14.63)**	**36 (29.27)**	**50 (40.65)**
I help children learn letters						
Anganwadi	1(1.59)	1(1.59)	3(4.76)	8(12.69)	18(28.57)	32(50.79)
Private	1(1.67)	2(3.33)	8(13.33)	5(8.33)	19(31.67)	25(41.67)
Total	**2 (1.63)**	**3 (2.44)**	**11 (8.94)**	**13 (10.57)**	**37 (30.1)**	**57 (46.34)**
I talk about the relationship between letters and sounds						
Anganwadi	2(3.17)	6(9.52)	5(7.94)	7(11.11)	11(17.46)	32(50.79)
Private	0	2(3.33)	7(11.67)	9(15)	16(26.67)	26 (43.33)
Total	**2 (1.63)**	**7 (5.7)**	**12 (9.75)**	**16 (13.01)**	**27 (21.95)**	**58 (47.15)**
I help children write						
Anganwadi	0	0	4(6.35)	2(3.17)	15(23.81)	42(66.67)
Private	0	0	3(5)	5(8.33)	17(28.33)	36(60)
Total	**0**	**0**	**7 (5.69)**	**7 (5.69)**	**32 (26.02)**	**78 (63.41)**
I am involved with reading and writing in play (that is setting up a doctor's clinic, kitchen set or write a grocery shop list where children can pretend to play)						
Anganwadi	0	1(1.58)	5(7.94)	7(11.11)	19(30.16)	31(49.21)
Private	1(1.67)	3(5)	4(6.67)	9 (15)	23(38.33)	20(33.33)
Total	**1 (0.81)**	**4 (3.25)**	**9 (7.32)**	**16 (13.01)**	**42 (34.15)**	**51 (41.46)**
I do alphabet activities with children (that is play with letter blocks, magnets, puzzles, use akshara grids, dice, word cards, ganitmala, blocks, beads, puzzles and board games or have imaginary play with dolls, masks, puppets, etc.)						
Anganwadi	0	0	6(9.52)	3(4.76)	15(23.81)	38(60.32)
Private	0	1(1.67)	0	8(13.33)	16(26.67)	35(58.33)
Total	**0**	**1 (0.81)**	**6 (4.88)**	**11 (8.94)**	**31 (25.2)**	**73 (59.35)**

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Discussion

The present study compared the self-reported emergent literacy beliefs and practices of preschool teachers in Anganwadi or private school types within the Udupi district in the South Indian context. This is critical since teachers' implementation of evidence-based emergent literacy practices is linked to later academic outcomes, which is noted as a focal point for the National Education Policy 2020 in India. Considering the fact that the two school types differ in emergent literacy curriculum and preschool teachers' perspectives on emergent literacy differ, understanding beliefs and practices may contribute to the development of portable and flexible teacher training in targeted school types or regions of India. Results indicated consistency with the use of effective developmentally appropriate practices identified as evidence-based practice, with few differences between the teachers in Anganwadi and private school types.

Teacher's self-reported beliefs about emergent literacy

Overall, responses indicated that teachers from both school types believed that emergent literacy is critical in the preschool years. Teachers responded most strongly that they play a major role in motivating and preparing children to read and write their names. They also strongly believed that their role involved monitoring and informing parents regarding children's literacy development and suggesting literacy activities, as well as asking children to choose books and discuss stories. Of the items related to the role of the preschool, teachers believed that preschools play a major role in encouraging early literacy in the preschool period, with a need to make provisions for a literacy-rich environment through play. Their beliefs on early literacy were consistent with the existing research in that teachers agreed on the importance of parents, the responsibility of schools and code-related skills (rhyming, phonics, print knowledge, alphabetic principle) with a play-based method to stimulate early literacy in children (Anjana and Rao 2019; Lopes and Fernandes 2009; Sandvik, Van Daal, and Adèr 2014). They also believed that children learn to spell in the same natural way that they acquire oral language skills, and children who enter school, able to recognise the letters of the alphabet, will learn to read more easily. Teachers' belief in developmentally appropriate practices is positively linked to their self-reports of the frequency of engagement in related instructional activities (Charlesworth et al. 1993; McMullen et al. 2005).

While teachers overall had strong beliefs, the total mean belief score for Anganwadi teachers was higher than that of private school teachers for the subsection of 'role of preschool teacher', which may have contributed to the significant difference in the total scores. One item that contributed to this difference was that Anganwadi teachers believed more strongly than private school teachers that they play a major role in implementing activities to develop conceptual skills (e.g. guiding children through storybooks, literacy in play). This finding is also similar to past research where some early educators are noted to adhere more to developmentally appropriate practices (Burts et al. 2000; Hegde and Cassidy 2009). The difference in the curriculum followed in the school contexts could have also influenced the teachers' beliefs towards emergent literacy.

Teachers' beliefs were not found to be influenced by teachers' experience level or their age. However, the results of this study indicated that teachers having higher qualifications

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had stronger beliefs regarding the role of the preschool in enhancing emergent literacy. This is not surprising since a higher level of teacher education is linked to better classroom quality and more developmentally appropriate practices (e.g. Lopes and Fernandes 2009). These results are somewhat different from other studies in Indian contexts that did not report educational effects (Anjana and Rao 2019; Hegde and Cassidy 2009). Anjana and Rao (2019) suggested that the provision of in-service training within the system may mitigate against education level effects.

Teacher's self-reported practices about emergent literacy

Overall, teachers' responses indicated consistency with the use of effective developmentally appropriate practices consistent with best-practice and prior cross-cultural research (Burts et al. 2000; Hegde and Cassidy 2009; Khurana and Rao 2011; Lopes and Fernandes 2009). Shared book reading emerged as a central practice, consistent with Khurana and Rao's (2011) finding that storybook reading to enhance oral language and print awareness was nearly universal in Kannada schools.

Of note in this study, private school teachers reported greater use of strategies related to concepts of books and print, and phonological awareness, whereas Anganwadi teachers relied more on general book reading routines. This difference may reflect disparities in training, curricular expectations or access to resources, yet both groups reported the use of book reading activities to a much higher degree than other practices such as concepts of books and print, phonological awareness and letter/*Akshara* knowledge. These activities included selecting developmentally appropriate books, naming things in pictures, connecting pictures to text, discussing unfamiliar words and emotions in stories, using voice modulation while reading, asking why and open-ended questions, using predictions, connecting to personal experiences and recalling stories. These practices suggest that teachers recognise the value of dialogic reading in supporting emergent literacy. However, the limited attention to phonics and phonological awareness (e.g. letter knowledge, sound-symbol correspondence, phoneme segmentation, alliteration) is notable. This gap may indicate a lack of emphasis in training or a preference for meaning-based over code-based approaches.

The teachers with higher qualifications focused more on practices such as shared book reading, the concept of book and print, literacy in play and phonological awareness than teachers with lower qualifications. Interestingly, this is different from the results from the survey regarding beliefs in that teachers had strong beliefs regardless of qualifications. It may be that highly qualified professionals obtained adequate training in the application of curriculum and strategies. In other words, all teachers may believe in the importance of emergent literacy, but do not have sufficient training to implement the practices (Lopes and Fernandes 2009; Lynch and Owston 2015; Sandvik, Van Daal, and Adèr 2014). In the Indian context, previous research by Hegde and Cassidy (2009) also suggested that the teachers with higher qualifications (bachelor's and above qualification level) focused on developmentally appropriate practices rather than rote learning, with the stated practices being congruent with the actual practices followed in the classroom. These findings call for frequent in-service training for teachers focusing on emergent literacy activities.

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In the present study, regardless of their years of experience, educational background, school setup or age, teachers spent approximately 15–20 min per day engaging them in various literacy activities. For some activities, they spent over 20 min (e.g. doing alphabet activities, reading and writing activities, talking about the relationship between letters and sounds, and story mapping). These results are consistent with other reports in Indian pre-primary schools. Zahedi et al. (2022) reported positive outcomes following a literacy programme that occurred five times a week for 30 min per class. Shenoy, Iyer, and Zahedi (2024) study reported that kindergarten children in a private middle-income school context are exposed to systematic phonics-based instruction for 30 min per day, for 5 days a week.

Emergent literacy in the multilingual environment

Although multilingualism was not the primary focus of this study, it represents an essential contextual factor in understanding emergent literacy practices in Indian preschools. Children in these classrooms are exposed not only to their home language but also to regional, national and instructional languages, which creates a more complex literacy environment than in monolingual contexts (Khurana and Rao 2011). Mohan, Francis, and Bhat (2022) found that exposure to native language script from a young age facilitated early literacy performance in a second language (English) among 140 typically developing bilingual preschool children. Superior performance in print skills in English was noted among preschool children having Malayalam as their native language due to early orthographic exposure, as compared to children having Tulu as their native language, who did not have a formal literacy experience or early orthographic exposure.

The Anganwadi schools follow Kannada predominantly, and activities for learning are carried out in the native language (Kannada). In this study, teachers in both school types reported the provision of *Akshara* activities that improved their letter recognition and encouraged them to create simple words. They used activities to teach letters of the alphabet, explained the differences between letter names and sounds, and encouraged children to write on slates, books and sand. In Kannada-speaking children, aged 5–10 years, learning to spell and read is reported to be slower (Nag 2007). Children in fourth and fifth grade are noted to have difficulty constructing complex *Akshara* using the consonant and vowel segments (Nag, Treiman, and Snowling 2010). Therefore, introducing phoneme and letter/*Akshara* knowledge at a younger age may support future multilingual literacy learning. However, the survey indicates that practices focused on letter/*Akshara* knowledge, phonological awareness and concepts of books and prints are notably lower than other activities.

In addition, some preschool teachers reported the use of *Shalapoorva Shiksha* and *Nali-Kali* (joyful learning/learning through playing) programmes. These child-centred, developmentally appropriate, activity-based teaching-learning programmes ensure self-paced, self-initiated learning for the children. The indigenously developed Nali-Kali programme involves active teacher–student participation in class. Specialised learning materials such as specific learning cards, flashcards, dice, beads, pebbles, sticks, puppets and screens are used, and experiential learning involves songs, games, workbooks and storytelling activities (Padmaja 2013; Raj et al. 2015).

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Implications for school programmes

The findings of the study indicate that teachers in both school types believe the importance of emergent literacy in preschool classrooms and believe that they play a role in the implementation of emergent literacy practices, including conceptual and procedural skill development. The findings point to some recommendations for in-service programmes, which were identified as critical in Indian contexts (Anjana and Rao 2019) and highlighted in NEP (2020).

First, although teachers in both school types responded with beliefs and practices consistent with evidence-based practice, in-service trainings that extend knowledge and application of both procedural and conceptual skills in both school types (such as letter/*Akshara* knowledge, phonological awareness and concepts of books and prints) may be beneficial. For example, training that leverages teachers' use of storybooks to target concepts of print (e.g. reading left to right, talking about the title, author and illustrator) may be helpful. In-service training could include content related to explicit training of phonics and/or phonological awareness skills. Trainings that involve practice and feedback sessions may provide additional support for teachers with lower qualifications (Anjana and Rao 2019; Girolametto, Weitzman, and Greenberg 2006).

Second, in-service trainings that focus on parent involvement to target language and literacy may be beneficial (Shenoy, Iyer, and Zahedi 2024; Zahedi et al. 2022). In particular, content related to the multilingual environment and inclusion of culturally diverse practices may be beneficial. Previous research has reported that parents of preschool children often engage in oral storytelling and shared book reading practices (Jacob et al. 2024; Pandith et al. 2022). Parental training regarding effective implementation of evidence-based strategies during these activities may be an additional boost to language and literacy development in preschool age.

Third, in the Indian school context, the use of a multilingual early literacy environment across school types may ensure that all children feel included in the learning process. There is a need to encourage the use of multilingual print in the classroom for labelling different areas in the school, for books, stories and poems (Menon 2020). Teachers from both school types included letter/*Akshara* knowledge. As such, it may be beneficial for teacher training to emphasise approaches concurrent with training in phonological awareness, considering the language diversity of the Indian context.

Limitations and future research

This study examined the self-reported literacy beliefs and practices of preschool teachers from two typical school setups. Overall, the study findings shed light on preschool teachers' perspectives regarding emergent literacy skills in the Indian context. The factors influencing teachers' beliefs and practices, and the extent to which emergent literacy exposure is prioritised before formal schooling in children is documented. However, the limitations hinder the generalisation of these findings, and future studies are needed. The study participants included teachers who began their teaching service during the COVID-19 pandemic, which could have prevented them from implementing the learning approaches. Although this study asked teachers to share their practices in an effort to give contents to the survey findings, data regarding actual classroom practices

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and reported practices are not obtained. Classroom observations were not conducted to verify the teachers' self-report. The initial questionnaire was not piloted with a group of early educators from each school type before administration and was only restricted to an expert review of the questionnaire. Further, the variation in the socio-economic status of the school types could have influenced the study findings.

Future research should examine parents' and school authorities' perspectives towards emergent literacy and explore effective classroom emergent literacy practices. Future research should investigate the degree of parent involvement and how teachers promote parental involvement to support school success. Understanding these practices may help identify ways parents can actively be involved in their child's learning activities, which is critical in multilingual contexts to bridge the home–school language gap (Khurana and Rao 2008). In addition, classroom observations of emergent literacy practices in multilingual contexts also need to be carried out in the Indian context to understand the best practices for teaching pre-requisite skills for reading development in multilingual contexts. Future studies also need to probe the cultural differences, if any, in the emergent literacy engagement across different school types. A systematic analysis of teacher focus groups from different types of schools can provide further insight into effective preschool practices.

It may be beneficial to employ emergent literacy activities throughout the day, not just while engaging in targeted activities. As such, it remains a question regarding the dosage of these activities (e.g. time, frequency per week). More research is also needed to understand the differential role and impact technology plays across school settings. Finally, more research is needed into the type and amount of in-service needed to support literacy goals and achievement in India.

Author contributions

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